All-as-Vision: Transferable Visual Exercise Representation for Knowledge Tracing

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Abstract

Knowledge tracing (KT) has long been built upon discrete question or concept identities, yet this representational habit sits uneasily with modern educational content, which is often visually structured, semantically rich, and continually refreshed. When a new exercise appears, an ID token carries almost no transferable prior; what matters is whether the model can encode the exercise itself as a semantic object. In this paper, we present *All-as-Vision*, a unified visual KT paradigm that renders heterogeneous educational content, including natural language, mathematical notation, tables, and diagrams, onto a shared image-based canvas and traces student knowledge directly in that common semantic space. The framework couples frozen visual embeddings with a lightweight causal gated backbone, thereby shifting the burden of generalization from sparse identity lookup to transferable content representation. Across standard KT benchmarks, All-as-Vision remains broadly competitive and attains leading results on large-scale settings such as Junyi and XES3G5M. Under stringent unseen-concept evaluation, the advantage becomes especially pronounced: on XES3G5M, visual tracing improves AUC from 0.9182 to **0.9283**, while on fully first-occurrence NIPS34 it increases AUC from 0.7459 to **0.8319**. Representation analysis further shows that visual embeddings induce neighborhoods that are more behaviorally coherent and more concept-sensitive than those produced by qid embeddings. Moreover, the same visual formulation supports direct cross-dataset transfer across all completed source-target pairs, reaching **0.7223** AUC from NIPS34 to XES3G5M and remaining viable even across substantially different benchmarks. Taken together, these results suggest that stronger KT generalization arises less from ever-heavier temporal machinery than from replacing sparse symbolic identities with richer and more portable exercise semantics.

CCS Concepts

• **Applied computing → Interactive learning environments**; • **Computing methodologies → Machine learning**.

Keywords

Knowledge Tracing, Visual Representation, Semantic Generalization, Educational Data Mining

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1 Introduction

Knowledge tracing (KT) aims to infer a student's evolving mastery state from historical interactions and to anticipate future performance. Over the past decade, the field has advanced from recurrent formulations to increasingly powerful attention-based architectures, steadily enriching how temporal dependency is modeled. Yet amid this architectural ascent, one foundational choice has remained comparatively underexamined: how the exercise itself is represented. In most existing KT systems, each question enters the model chiefly as a discrete identifier. Such a representation is serviceable when the item bank is stable and densely observed, but it reduces an exercise to an arbitrary address and leaves its semantic substance outside the model's field of view.

This representational habit becomes increasingly brittle on modern educational platforms. Real item banks are continuously refreshed, and many exercises are intrinsically multimodal, weaving together natural language, mathematical notation, tables, symbolic layout, and diagrams. For newly introduced items, a qid carries no transferable prior at all; it merely marks novelty without explaining meaning. In such a regime, the core limitation is not that sequence backbones are too weak, but that the representational substrate is too sparse precisely where generalization is most needed.

In this paper, we propose **All-as-Vision**[1], a content-first KT paradigm in which heterogeneous exercise materials are rendered onto a shared canvas and encoded by a pretrained visual encoder. Rather than treating an exercise as an opaque symbolic token, All-as-Vision treats it as a semantic object whose wording, notation, spatial arrangement, and diagrammatic cues can all be absorbed within a unified visual interface. The resulting embeddings are then traced by a lightweight causal gated backbone, so that temporal modeling is built on top of a semantically grounded exercise space rather than a sparse identity table.

Recent text-aware KT methods already point in this direction by injecting textual semantics into question representation [14, 15, 25]. However, most of them still fuse content back into an ID-centered pipeline, which leaves a shortcut path through seen question identities and makes it difficult to separate semantic transfer from memorization. Moreover, many educational items are not losslessly serializable: mathematical layout, geometric diagrams, and table structure often carry essential semantics. We therefore do not position All-as-Vision as a claim that vision is categorically superior to language. Rather, our claim is that a unified pixel-level interface offers

[1]Anonymous repository: https://anonymous.4open.science/r/all-as-vision-3B62.

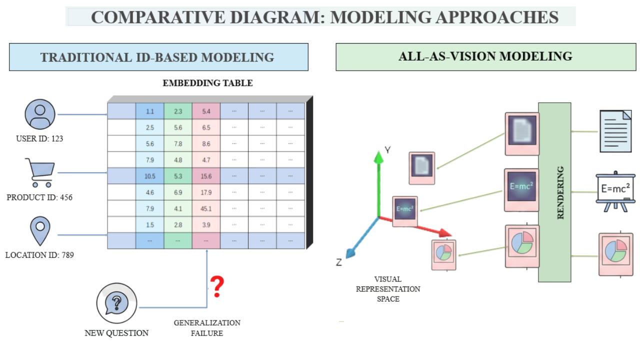

Figure 1: Comparison between traditional ID-based KT and the proposed All-as-Vision approach. Left: question-ID-based tracing provides no semantic signal for exercises not seen during training. Right: heterogeneous educational content is encoded into a shared visual representation space, which a causal tracing backbone then operates over.

a zero-engineering route for absorbing heterogeneous educational content without dataset-specific multimodal alignment.

To test whether this representational pivot yields practical consequences, we evaluate the method along three increasingly demanding axes. First, under conventional KT benchmarking, All-as-Vision remains broadly competitive and already reaches leading results on large-scale settings such as Junyi and XES3G5M. Second, under strict unseen-question and unseen-concept evaluation, where identifier reuse can no longer rescue generalization, the visual formulation becomes decisively stronger, culminating in a large gain on fully first-occurrence NIPS34. Third, we show that the same representation can transfer directly across datasets, a setting in which conventional qid-based KT models are not merely weak but structurally inapplicable. We further support this empirical picture with representation-space analysis and a theory-grounded argument that visual semantics reduce the effective hypothesis complexity of the downstream tracer. The contributions are as follows:

- We propose **All-as-Vision**, a unified visual KT paradigm that renders heterogeneous educational exercises into a common canvas, encodes them with a frozen vision backbone, and performs tracing over the resulting semantic representation space.
- We establish the paradigm's value across **standard benchmarking, unseen-concept evaluation, and cross-dataset transfer**, showing that visual exercise representations remain competitive in conventional KT while becoming markedly stronger when semantic generalization is truly required.
- We provide **structural evidence and theoretical support** through behavior-coherence analysis, concept-neighborhood analysis, and a hypothesis-space account of why semantically organized exercise spaces ease downstream learning.

2 Related Work

2.1 KT and Exercise-aware Representation

KT has evolved from classical probabilistic formulations to increasingly expressive neural sequence models. DKT [20] established recurrent tracing as a viable paradigm, while later models such as DKVMN [30], KQN [10], SAKT [19], AKT [3], SAINT, and SAINT+ [2, 22] strengthened memory access, interaction matching, and long-range dependency modeling. Across these lines, however, the exercise is still usually introduced through a discrete qid, occasionally accompanied by concept tags or lightweight metadata, so the burden of generalization is pushed onto the temporal backbone.

Recent KT research has therefore begun to enrich exercise representation with content information. EKT [14] explicitly models exercise-aware signals, while question-embedding pretraining approaches [15, 25] use textual semantics to strengthen question representations before tracing. These works are important precursors, yet most still reinject content into an ID-oriented pipeline, leaving room for shortcut learning through seen identities and rarely covering the broader visual structure of educational materials. What remains underdeveloped is a genuinely content-first KT formulation in which the exercise itself becomes the primary representational object.

2.2 From Symbolic IDs to Transferable Semantic Representations

The move away from symbolic identifiers toward dense content-native representations is not unique to KT; it reflects a broader evolution across modern deep learning. Earlier systems often relied on learned embeddings tied to dataset-specific symbols, whereas more recent approaches increasingly encode the content itself, allowing semantically related instances to occupy nearby regions in representation space. In language and multimodal learning alike, this shift has produced representations that transfer more naturally across tasks, domains, and even modalities.

This trend becomes especially visible with large pretrained alignment models such as CLIP [21], SigLIP [28], and SigLIP 2 [23], which organize visual inputs according to semantic compatibility rather than task-local identifiers. Subsequent analyses also suggest that such objectives induce meaningful global geometry rather than a purely arbitrary embedding cloud [9, 12]. Recent work on optical context compression and visual-language pretraining over structured screenshots or documents, such as DeepSeek-OCR [27], Pix2Struct [11], and Donut [8], further echoes this broader intuition by treating textual content itself as a structured visual interface rather than as a purely sequential token stream. From this perspective, representing an educational exercise purely by qid appears increasingly outdated: an exercise is not merely a database key, but a semantic artifact composed of wording, notation, layout, and often diagrammatic relations.

This broader representational movement is particularly relevant to education, where meaning is frequently borne by visual form itself. Mathematical expressions are spatially organized, geometry is irreducibly diagrammatic, and many assessment items depend on alignment, tabular structure, or mixed symbolic-textual composition. Consequently, the issue is not merely that KT needs "more features" attached to qids; rather, it needs a representational pivot toward a transferable semantic space in which heterogeneous exercise content can be encoded natively. Our work instantiates precisely this pivot by treating all exercise content as visual input and performing KT directly over that shared semantic manifold.

Table 1: Key notations used in All-as-Vision.

Notation	Description
$\mathcal{S}_u$	Historical interaction sequence of a student
q_t, r_t	Exercise content and correctness label at step t
I_t	Rendered visual canvas of q_t
$\mathcal{R}(\cdot)$	Rendering operator
$\mathbf{v}_t$	Visual semantic embedding of q_t
$\mathbf{e}(r_t)$	Response embedding for r_t
$\mathbf{z}_t$	Interaction representation $[\mathbf{v}_t; \mathbf{e}(r_t)]$
$\mathbf{h}_t$	Latent student state after processing the prefix up to t
$q_{T+1}, \hat{r}_{T+1}$	Target exercise and predicted correctness probability
$\mathcal{L}$	Binary cross-entropy training objective

3 Problem Setup and Notation

We consider the standard next-response prediction problem in KT, but alter the representational basis on which the problem is solved. For a student u, the historical interaction sequence is written as

$$\mathcal{S}_u = \{(q_1, r_1), (q_2, r_2), \ldots, (q_T, r_T)\}, \tag{1}$$

where q_t denotes the raw exercise content presented at step t and $r_t \in \{0, 1\}$ denotes whether the student answers that exercise correctly. Unlike conventional KT, where an exercise is often reduced at the outset to a discrete question ID or concept ID, we treat q_t as a content-rich object that may contain natural language, formulas, tables, diagrams, or mixed layouts.

To make the computational pipeline explicit, we distinguish four successive representational levels. First, raw exercise content is converted into a rendered visual canvas,

$$I_t = \mathcal{R}(q_t), \tag{2}$$

where $\mathcal{R}(\cdot)$ denotes the rendering operator. Second, the canvas is encoded into a dense semantic vector,

$$\mathbf{v}_t = \mathrm{Enc}(I_t). \tag{3}$$

Third, the exercise semantics are combined with the observed response to form an interaction representation $\mathbf{z}_t$. Finally, the interaction sequence is accumulated into a latent student state $\mathbf{h}_t$. Given a target exercise q_{T+1}, the predictive goal is to estimate

$$\hat{r}_{T+1} = P(r_{T+1} = 1 \mid \mathcal{S}_u, q_{T+1}). \tag{4}$$

Table 1 summarizes the principal notations.

4 Methodology

4.1 Unified Visual Exercise Representation

The first stage of All-as-Vision constructs a unified visual interface for heterogeneous educational exercises. Educational questions are frequently multimodal in substance even when they appear text-centric on the surface: line breaks, formula layout, answer formatting, tables, and embedded diagrams often carry part of the pedagogical meaning. If such content is prematurely collapsed into sparse IDs or aggressively linearized symbolic strings, substantial educational structure is discarded before tracing even begins.

We therefore render each exercise into a two-dimensional canvas,

$$I_t = \mathcal{R}(q_t), \tag{5}$$

that preserves its original spatial organization. Text-only items are formatted into clean image-like layouts, formula-rich exercises retain their native two-dimensional notation, and diagram-containing problems remain visual rather than being reduced to auxiliary tags. The purpose of this step is not merely to turn everything into images, but to create a common representational interface across datasets with very different exercise formats.

To make this step reproducible, we use a standardized rendering pipeline rather than dataset-specific manual templates. Text and formulas are composed onto a white canvas with layout-aware formatting, while native figures or diagrams are preserved in their original relative positions whenever available. All canvases are then normalized to the encoder input size with aspect-ratio-preserving scaling and white padding, rather than naive stretching or cropping, so that mathematical layout and geometric relations are not inadvertently distorted. This design is important because, in educational content, small geometric or typographic deformations can alter semantics rather than merely appearance.

The rendered canvas is then mapped into a dense visual semantic embedding

$$\mathbf{v}_t = \mathrm{Enc}(I_t), \tag{6}$$

where $\mathrm{Enc}(\cdot)$ is a pretrained vision backbone such as SigLIP [28] or SigLIP 2 [23]. We keep the encoder frozen during KT training so that the KT model inherits a denser and more transferable exercise space than qid lookup can offer. A newly introduced exercise may be unseen as an identity, yet it can still be embedded near semantically related items immediately after rendering. The output of this stage is the vector $\mathbf{v}_t$, which serves as the semantic anchor for all subsequent tracing operations.

4.2 Interaction Encoding and Causal State Update

Once each exercise has been placed in the shared visual semantic space, the next question is how to incorporate student behavior. In KT, the model must represent not only what semantic content the student encountered, but also whether that encounter revealed success or difficulty. We therefore bind together exercise semantics and learner outcome through an interaction representation:

$$\mathbf{z}_t = [\mathbf{v}_t; \mathbf{e}(r_t)]. \tag{7}$$

Here $\mathbf{e}(r_t)$ denotes the response embedding associated with the binary correctness signal. The resulting vector $\mathbf{z}_t$ is a compact event-level summary of what the student saw and how the student responded to it. This interaction representation is then used to update a latent student state through a lightweight gated causal backbone:

$$\mathbf{i}_t = \sigma(\mathbf{W}_i \mathbf{z}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i), \tag{8}$$

$$\mathbf{f}_t = \sigma(\mathbf{W}_f \mathbf{z}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f), \tag{9}$$

$$\mathbf{h}_t = \mathbf{f}_t \odot \mathbf{h}_{t-1} + \mathbf{i}_t \odot \mathrm{SiLU}(\mathbf{W}_v \mathbf{v}_t + \mathbf{b}_v), \tag{10}$$

where σ is the sigmoid function and $\odot$ denotes element-wise multiplication. Intuitively, the input gate $\mathbf{i}_t$ controls how strongly the current interaction should modify the student's latent state, while the forget gate $\mathbf{f}_t$ controls how much previously accumulated knowledge should be retained. The resulting state $\mathbf{h}_t$ summarizes what the model currently believes the student knows after processing the prefix up to time t.

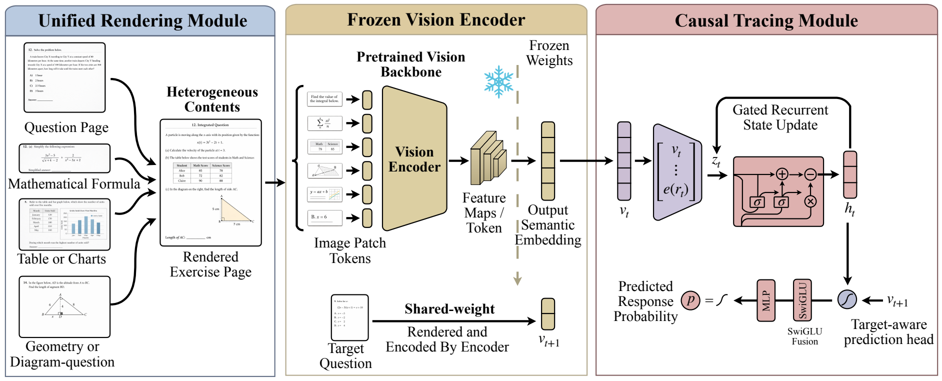

Figure 2: Overview of the All-as-Vision pipeline. Heterogeneous educational exercises are first rendered into a unified visual format, encoded into frozen visual embeddings, processed by a lightweight causal tracing backbone, and finally combined with the target exercise embedding for next-response prediction.

This update is strictly causal: $\mathbf{h}_t$ depends only on the observed prefix and never on future interactions. More importantly, the backbone is intentionally lightweight. Our claim is that a sufficiently rich exercise manifold can reduce the representational burden placed on the sequence model itself.

This design choice is central rather than incidental. In conventional KT, the backbone must simultaneously memorize sparse item identities, infer latent relatedness among exercises, and propagate student evidence through time. All-as-Vision deliberately separates these roles. The frozen visual encoder is asked to supply the semantic organization of the exercise space in advance, while the causal gated tracer only needs to model how correctness signals should accumulate, persist, or fade over that already structured space. In that sense, the backbone is not weakened for convenience; it is simplified because the representation is expected to carry more of the semantic burden from the outset.

4.3 Target-aware Prediction and Training Objective

After processing the historical prefix, the model must evaluate the target exercise against the student's current latent mastery. To do so, the target item q_{T+1} is rendered and encoded in exactly the same way as the historical exercises, yielding the target semantic vector $\mathbf{v}_{T+1}$. Prediction is then performed by combining the final student state with the target embedding:

$$\mathbf{e}_{T+1} = \mathrm{SwiGLU}(\mathbf{W}_e[\mathbf{h}_T; \mathbf{v}_{T+1}] + \mathbf{b}_e), \tag{11}$$

$$\hat{r}_{T+1} = \sigma(\mathrm{MLP}(\mathbf{e}_{T+1})). \tag{12}$$

This step is target-aware in a precise sense. The student state alone cannot specify which competencies are being queried, while the target exercise alone cannot reveal whether the student has mastered them. For each valid next-step target, the model is optimized with binary cross-entropy:

$$\mathcal{L} = -\sum_{u}\sum_{t=1}^{T_u-1}\Big(r_{t+1}\log\hat{r}_{t+1} + (1-r_{t+1})\log(1-\hat{r}_{t+1})\Big). \tag{13}$$

This objective is standard in form, but it supervises a different computational pathway from ordinary ID-based KT. Instead of learning primarily how to memorize item identities, the model learns how student state should evolve over a semantically structured exercise manifold. Algorithm 1 summarizes the complete forward procedure.

Algorithm 1 All-as-Vision Forward Procedure

Require: Student prefix $\mathcal{S}_u = \{(q_t, r_t)\}_{t=1}^{T}$ and target exercise q_{T+1}
Initialize latent state $\mathbf{h}_0 \leftarrow \mathbf{0}$
for $t = 1$ to T **do**
  Render the exercise canvas: $I_t \leftarrow \mathcal{R}(q_t)$
  Extract frozen semantic features: $\mathbf{v}_t \leftarrow \mathrm{Enc}(I_t)$
  Form interaction summary: $\mathbf{z}_t \leftarrow [\mathbf{v}_t; \mathbf{e}(r_t)]$
  Update latent student state via gated transition to obtain $\mathbf{h}_t$
end for
Render and encode the target exercise: $I_{T+1} \leftarrow \mathcal{R}(q_{T+1})$, $\mathbf{v}_{T+1} \leftarrow \mathrm{Enc}(I_{T+1})$
Fuse current mastery with target semantics: $\mathbf{e}_{T+1} \leftarrow \mathrm{SwiGLU}(\mathbf{W}_e[\mathbf{h}_T; \mathbf{v}_{T+1}] + \mathbf{b}_e)$
Predict next-step correctness: $\hat{r}_{T+1} \leftarrow \sigma(\mathrm{MLP}(\mathbf{e}_{T+1}))$
Optimize the binary cross-entropy loss over all training targets

The methodology can therefore be read as a single coherent pipeline: All-as-Vision first constructs a unified visual view of heterogeneous educational content, then places exercises into a transferable semantic space, then accumulates student behavior causally over that space, and finally predicts future correctness by matching the resulting student state against the semantics of the target exercise.

It is also important to distinguish offline preprocessing from online inference. The expensive part of the pipeline, namely rendering exercises and extracting SigLIP embeddings, is performed strictly offline when a question enters the item bank. In deployment, these visual embeddings are cached once and retrieved in $O(1)$ time during student tracing. The online stage therefore consists only of vector lookup followed by our lightweight gated update and target-conditioned prediction. In this sense, All-as-Vision shifts complexity from repeated online inference to one-time offline preprocessing, while keeping online tracing fully comparable to standard KT systems. In our measurement on an RTX 4090D, SigLIP2-Large-384 requires 7.77 ms/image for offline encoding, whereas the cached online tracer runs at 0.1522 μs/token with 9.19M parameters, making it about 6.1× faster than DKVMN, 7.1× faster than simpleKT, and 14.9× faster than FoLiBiKT in normalized per-token latency. This separation is also attractive in practice for educational platforms, where item banks evolve far more slowly than student interaction streams.

The tracing module is deliberately minimal not because temporal modeling is unimportant, but because our design hypothesis assigns different roles to representation and sequence learning. The frozen visual encoder is responsible for organizing exercises into a semantically meaningful input manifold, while the recurrent gated tracer is responsible for accumulating student-specific evidence over that manifold. This division of labor is central to the method: if the exercise space already places related items near one another, then the temporal backbone no longer needs to reconstruct semantics from repeated identity exposure. Instead, it only needs to learn how success and failure signals should propagate across a structured neighborhood of exercises. In this sense, All-as-Vision does not aim to replace temporal KT modeling; it aims to place temporal KT modeling on a more transferable substrate from the start.

4.4 Representation-Induced Hypothesis Space Reduction

The discussion above also suggests a formal reason why strong performance can emerge even with a lightweight temporal backbone. Our perspective is inspired by theoretical analyses of contrastive and self-supervised representation learning, which connect structured representations to improved downstream transfer, geometric regularity, and more favorable sample complexity [1, 5, 24]. Let $q \in \mathcal{Q}$ denote a question and h a student interaction history. A KT predictor built on representation $\phi(\cdot)$ can be written as

$$g_\phi(q, h) = f(\phi(q), h), \tag{14}$$

where f is the downstream tracing model. Rather than studying f in isolation, we consider the *induced function class*

$$\mathcal{F}_\phi = \{(q, h) \mapsto f(\phi(q), h) \mid f \in \mathcal{G}\}, \tag{15}$$

where $\mathcal{G}$ is a capacity-controlled family of tracers. A standard generalization argument then yields

$$\mathcal{E}_{\text{gen}}(g_\phi) \le \mathcal{E}_{\text{train}}(g_\phi) + O\big(\mathfrak{R}_n(\mathcal{F}_\phi)\big), \tag{16}$$

so the critical question is not only how expressive f is, but also how the representation ϕ reshapes the effective complexity of the admissible predictor class.

Under ID-based KT, $\phi_{\text{id}}(q)$ treats each question as an independent symbol. Semantically related exercises may therefore remain orthogonal in representation space, leaving the downstream model free to assign unrelated behaviors to them. By contrast, pretrained visual encoders such as CLIP, SigLIP, and SigLIP 2 are learned through alignment objectives that place semantically matched content into a shared geometry [21, 23, 28]. Prior analyses further suggest that these objectives induce nontrivial structure, including alignment, uniformity, local smoothness, and global regularity, in the learned space [9, 12, 24]. For educational exercises, this means that wording, notation, layout, and diagrammatic similarity can already be partially organized before KT training begins.

We therefore focus on two data-dependent properties of the visual space that are directly testable in our experiments: *clustering structure*, where semantically related questions form compact local groups, and *local response consistency*, where nearby questions exhibit similar response behavior conditioned on history. When both properties hold, the predictor no longer needs a fully question-wise decision rule; it can be approximated by a lower-variance rule shared across local semantic neighborhoods.

More concretely, suppose the visual space admits a cover $\{\mathcal{N}_k\}_{k=1}^K$ of compact neighborhoods such that questions inside the same neighborhood are both semantically close and behaviorally similar. Then the predictor can be decomposed as $g_\phi(q, h) \approx \tilde{g}(c(q), h) + \Delta(q, h)$, where $c(q)$ indexes the neighborhood containing q and Δ is a small residual term. In the idealized qid case, by contrast, $c(q)$ degenerates into the question identity itself, so no compression is obtained. The crucial implication is that visual structure changes the granularity at which the tracer must fit behavior: from isolated symbols toward semantically coherent groups.

Proposition 1 (Representation-Induced Simplification). If a representation ϕ induces compact semantic neighborhoods together with locally consistent response behavior, then the effective complexity of $\mathcal{F}_\phi$ is reduced relative to an unstructured ID representation, because prediction can be compressed from question-level memorization toward neighborhood-level sharing.

Interpretation. Under ϕ_{id}, many arbitrary question-specific labelings remain admissible. Under a structured visual representation, however, one rule can cover a family of nearby questions whose semantics and behavior are already aligned. This removes many otherwise admissible but non-transferable solutions from the hypothesis space, which in turn tightens the downstream generalization requirement.

This viewpoint also clarifies why the empirical gains are strongest under strict generalization settings. When evaluation still contains dense question recurrence, an ID-based model can partially hide representational weakness through repeated lookup and memorization. Once evaluation shifts toward first-occurrence questions, unseen concepts, or unseen datasets, that shortcut collapses and representation geometry becomes dominant. This proposition is deliberately aligned with our empirical design: clustering and local response consistency are examined in RQ3, while the resulting reduction in effective complexity is expressed downstream in RQ2 and RQ4.

5 Experiments

5.1 Datasets and Protocols

Our experimental design serves two purposes: to position All-as-Vision as a general KT framework and to report the strongest verified evidence already in hand with clear scope control. Concretely, this section is organized around two components:

- **Datasets.** We frame the study around three standard KT benchmarks: **Junyi** [7], a large-scale mathematics practice log that provides question text, which we render into the shared canvas for visual encoding; **NIPS34** [26], derived from the NeurIPS 2020 Education Challenge and equipped with native question

Table 2: Dataset statistics.

Dataset	#Students	#Questions	#Concepts	#Interactions
Junyi	72,630	721	721	16,252,437
XES3G5M	18,066	7,652	865	5,549,635
NIPS34	4,918	948	57	1,382,727

images, making it especially suitable for strict semantic cold-start analysis; and **XES3G5M** [17], a recent large-scale benchmark with both question text and associated visual material, which we jointly compose on the same shared canvas before encoding.

- **Protocols.** We conduct three complementary experimental tracks. The first follows standard KT benchmarking practice: all methods are evaluated under a five-fold setting, monitored with early stopping, and selected by validation AUC after hyperparameter sweep. The second is an *unseen-concept* evaluation: we sample a subset of concepts and remove all questions associated with those concepts from the training side, so that the held-out split measures whether a model can transfer to genuinely unseen semantic targets rather than relying on repeated question identity. The third is a *cross-dataset transfer* setting, where a model trained on one benchmark is directly evaluated on another to test whether its representation space remains usable beyond a single dataset.

5.2 Baselines

The comparison spans representative KT families rather than a single architectural line. Specifically, we include recurrent baselines such as DKT [20] and DKT-Forget [18], memory-augmented models such as DKVMN [30], interaction-aware variants such as KQN [10] and ATKT [4], and attention-based tracers including SAKT [19], SAINT [2], AKT [3], simpleKT [16], stableKT [13], and FoLiBiKT [6].

Pure-text baselines are not included in the main benchmark for two reasons. First, key semantics in our benchmarks, especially formulas, layout, tables, and diagrams, are severely corrupted by text-only extraction. Second, existing text-aware KT methods such as EKT [14], pre-trained question embedding approaches [15], and SEEP [25] typically reinject semantic signals into an ID-centered pipeline, which leaves a shortcut path through seen identities and makes semantic transfer harder to disentangle from memorization. For reference, our auxiliary text baseline follows the same comparison philosophy as the qid ablation: it extracts BGE-M3 text embeddings and feeds them into the same causal tracing backbone used by All-as-Vision, rather than introducing a separate temporal architecture.

These three tracks serve distinct but complementary purposes. The standard benchmark verifies that All-as-Vision remains competitive under the conventional KT evaluation regime. The unseen-concept protocol asks whether the model can generalize when target questions and concepts are nearly entirely unseen. The cross-dataset transfer setting then pushes one step further by asking whether the learned representation can survive even when the source and target datasets themselves differ.

5.3 Research Questions

To make the experimental logic explicit, we organize the remainder of this section around four research questions.

- **RQ1:** Can All-as-Vision be established as a *general and leading* KT paradigm, rather than as a technique whose value is confined to a particular unseen-concept evaluation protocol?
- **RQ2:** For settings with almost entirely unseen questions and concepts, can All-as-Vision outperform existing KT models?
- **RQ3:** Do the gains of All-as-Vision coincide with a more semantically ordered and behaviorally coherent representation space?
- **RQ4:** Can All-as-Vision support cross-dataset transfer in a way that conventional ID-based KT models cannot?

5.3.1 RQ1: Generality Beyond Protocol-specific Holdout. Our answer is *yes.* The standard benchmark results already show that All-as-Vision is not merely a protocol-specific curiosity confined to unseen-concept stress tests. On Junyi, it delivers the strongest overall performance, reaching **0.8015** AUC and **0.8538** ACC. On XES3G5M, it again achieves the best AUC at **0.8633**, outperforming the strongest conventional baselines, including FoLiBiKT at 0.8554 and SAINT at 0.8523. On NIPS34, stableKT attains a slightly higher AUC under standard evaluation, while All-as-Vision remains tightly competitive at **0.8046**, essentially matching the strongest qid-based alternatives on a benchmark still favorable to identity recurrence. Taken together, these results show that the visual formulation does not sacrifice general KT capability under the conventional regime; on larger settings it already rises to the top, and on NIPS34 it remains within the narrow leading group.

5.3.2 RQ2: Performance Under First-occurrence Unseen-concept Evaluation. Our answer is *yes.* We instantiate this setting through a *pure concept-heldout* protocol. XES3G5M is already highly strict, while NIPS34 is fully first-occurrence, making it the clearest stress test in this paper.

The results in Table 5 are direct. Here we control the tracing architecture and only replace the question representation: the qid variant feeds learned ID embeddings into the same lightweight tracer that our visual model uses with frozen visual embeddings. Under this controlled comparison, All-as-Vision improves XES3G5M from 0.9182 to **0.9283**. On NIPS34, where first-occurrence ratio reaches 1.0, the gap expands from 0.7459 to **0.8319**. Once both questions and concepts are effectively unseen, the advantage of semantic visual representation becomes unequivocal.

5.3.3 RQ3: Embedding Geometry and Semantic Structure. Our answer is again *yes.* Figure 3 offers the visual intuition: qid neighborhoods remain heavily intermixed, whereas visual embeddings become markedly more compact and better separated.

These numbers make the geometric intuition operational. KT generalization is ultimately local: when a new question arrives, the model can only transfer from whatever neighbors the representation makes available. Under qid, those neighbors are largely arbitrary; under visual encoding, they are far more likely to share both semantic content and behavioral regularity. This is also consistent with broader evidence that contrastive objectives tend to support cleaner cluster structure and more useful neighborhoods for downstream tasks [29]. Figure 3 makes the contrast concrete:

Table 3: Main benchmark results under standard KT evaluation. Mean±std is reported when available; otherwise verified point estimates are shown. Best results are highlighted in bold.

Model	Junyi2015		NIPS task34		XES3G5M	
	AUC (%) ↑	ACC (%) ↑	AUC (%) ↑	ACC (%) ↑	AUC (%) ↑	ACC (%) ↑
DKT [20]	70.75±0.05	83.20±0.04	76.89±0.02	70.32±0.04	83.37±0.21	84.05±0.08
DKVMN [30]	74.83±0.76	84.53±0.18	76.73±0.04	70.16±0.05	82.86±0.10	83.85±0.04
KQN [10]	74.65±1.18	84.36±0.36	76.84±0.03	70.28±0.01	79.46±0.93	82.90±0.88
DKT-Forget [18]	75.56±0.06	84.65±0.10	77.73±0.03	70.76±0.02	57.55±0.08	66.54±0.04
ATKT [4]	73.24±0.06	84.17±0.03	76.65±0.01	70.13±0.02	79.97±0.03	83.89±0.05
SAKT [19]	70.62±0.15	83.78±0.24	75.17±0.05	68.79±0.04	80.92±0.73	83.65±0.65
SAINT [2]	71.53±0.28	83.65±0.34	78.73±0.07	71.80±0.06	85.23±0.76	85.02±0.38
AKT [3]	77.54±0.24	84.42±0.72	80.33±0.03	73.23±0.05	80.05±2.94	82.65±1.15
simpleKT [16]	<u>77.80</u>±0.21	<u>84.70</u>±0.08	80.35±0.00	73.28±0.01	78.92±3.72	83.58±1.05
stableKT [13]	77.37±0.06	84.60±0.04	**80.59**±0.04	**73.48**±0.05	85.36±0.20	85.28±0.20
FoLiBiKT [6]	76.63±0.20	84.20±0.25	80.33±0.02	73.23±0.01	<u>85.54</u>±0.62	**85.61±0.77**
All-as-Vision	**80.15**±0.05	**85.38**±0.12	<u>80.46</u>±0.06	<u>73.32</u>±0.05	**86.33±0.04**	<u>85.35</u>±0.05

Table 4: Protocol statistics under unseen-concept evaluation instantiated by pure concept-heldout splitting.

Metric	XES3G5M	NIPS34
Held-out concepts	214/858	14/57
Held-out questions	1401/7652	162/948
Held-out question ratio (%)	18.31	17.09
Training retain ratio (%)	83.99	84.80
First-occurrence ratio (%)	90.40	100.00
Prior same-question ratio (%)	9.60	0.00

Table 5: Verified results under first-occurrence unseen-concept evaluation.

Dataset	Model	AUC (%) ↑	ACC (%) ↑
XES3G5M	same tracer + qid	91.82	86.11
XES3G5M	All-as-Vision (visual)	**92.83**	**87.14**
NIPS34	same tracer + qid	74.59	70.88
NIPS34	All-as-Vision (visual)	**83.19**	**75.10**

Table 6: Behavioral coherence and concept structure on XES3G5M. Lower continuity is better; higher concept metrics are better.

Representation	Continuity (%) ↓	Same-concept NN (%) ↑	KMeans NMI (%) ↑
qid	81.50	0.50	57.26
visual	**79.67**	**49.46**	**68.15**

qid keeps algebraically or visually similar exercises apart unless supervision has repeatedly tied their identities together, whereas the visual space already brings such items into local proximity through

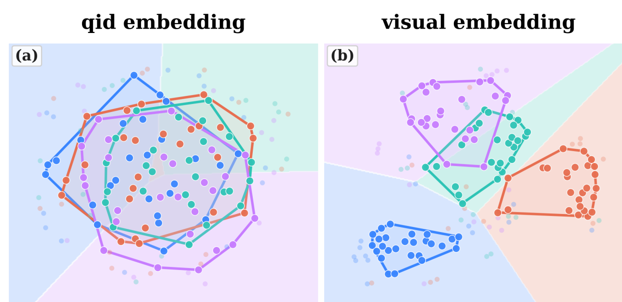

Figure 3: t-SNE visualization of qid and visual embedding geometries. In the qid space, concept groups remain heavily intermixed; in the visual space, they become markedly more compact and better separated.

layout, notation, and diagrammatic similarity. That geometric advantage is not cosmetic; it creates the neighborhood structure that strict generalization in RQ2 actually requires.

5.3.4 RQ4: Zero-shot Cross-dataset Transferability. Our answer is also *yes*. Conventional qid-based KT baselines are structurally excluded from this evaluation: their embedding tables are tied to a fixed, dataset-specific vocabulary of question identities, so zero-shot cross-dataset transfer without target-side identity remapping becomes mathematically ill-posed. Because All-as-Vision instead maps exercises into a shared visual semantic space, it supports this transfer mode in a way that conventional KT baselines fundamentally cannot.

Table 7 shows that this capability is systematic across all completed transfer directions. The strongest result arises from NIPS34 → XES3G5M, which attains **72.23 ± 0.70** AUC and 79.69±0.78 ACC. Transfer to Junyi remains viable from both NIPS34 and XES3G5M, while the reverse directions into NIPS34 are understandably harder. Even so, every completed direction remains operational without

Table 7: Zero-shot cross-dataset transfer results. Each model is trained on the source benchmark and directly evaluated on the target benchmark without target-side identity alignment.

Source → Target	AUC (%) ↑	ACC (%) ↑
NIPS34 → XES3G5M	**72.23 ± 0.70**	79.69 ± 0.78
NIPS34 → Junyi2015	**69.72 ± 0.85**	80.15 ± 2.61
XES3G5M → NIPS34	68.75 ± 1.68	62.99 ± 1.21
XES3G5M → Junyi2015	67.50 ± 1.01	**82.70 ± 0.71**
Junyi2015 → NIPS34	66.80 ± 1.49	60.85 ± 0.95
Junyi2015 → XES3G5M	67.82 ± 1.07	78.76 ± 0.41

Table 8: Ablation results on visual backbone scaling and post-processing for All-as-Vision on XES3G5M.

Setting	Variant	AUC (%) ↑
Backbone scaling	SigLIP1-Base-224	85.24
Backbone scaling	SigLIP2-Base-224	85.52
Backbone scaling	SigLIP2-Large-384	**86.33**
Encoder variant	SigLIP2-Base-NaFlex	85.41
Post-processing	default	**86.33**
Post-processing	l2	85.60
Post-processing	mean	85.43
Post-processing	gem	85.07

target-side identity remapping, highlighting the portability of the visual interface across datasets.

This result should be read as more than another transfer number. In standard KT pipelines, question identity is not merely one feature among others; it is the indexing scheme of the entire model. Once that indexing scheme changes, the model loses its basic interface to the target benchmark. All-as-Vision avoids this failure mode by grounding transfer in rendered content rather than dataset-local symbols. The point is therefore not only whether parameters trained on one benchmark happen to transfer to another, but whether the representational interface itself survives the benchmark shift. Under qid-based KT, it does not; under All-as-Vision, the interface remains the rendered exercise itself. That distinction is exactly what makes zero-shot cross-dataset transfer meaningful here rather than ill-posed from the outset.

5.3.5 Additional Ablation: Backbone Scaling and Post-processing. We further examine whether the gains mainly come from scaling the frozen visual backbone itself or from extra post-processing.

Table 8 yields two direct observations. First, scaling the frozen visual backbone consistently improves AUC, culminating in **86.33**% with SigLIP2-Large-384. Second, once the strongest backbone is fixed, alternative post-processing variants all underperform the default pipeline. The main gain therefore comes from backbone quality rather than post-hoc engineering.

5.3.6 Optimization-side Evidence: Best-checkpoint Epochs. We also examine whether the visual representation reduces optimization burden in practice. Using the validation-optimal epoch as a compact proxy, Table 9 shows that All-as-Vision typically reaches a high-quality solution earlier than several strong qid-based baselines.

Table 9: Best-checkpoint epochs on Junyi2015 and XES3G5M. 'Best AUC' denotes the best validation AUC. Smaller epochs indicate that the validation-optimal checkpoint is reached earlier.

Dataset	Model	Best AUC	Epoch
Junyi2015	*All-as-Vision*	**0.8047**	6
	simpleKT	0.7858	43
	stableKT	0.7801	17
	DKVMN	0.7536	13
	KQN	0.7534	16
	DKT+	0.7468	8
XES3G5M	*All-as-Vision*	**0.8628**	9
	stableKT	0.8603	43
	SAINT	0.8569	16
	simpleKT	0.8540	20
	DKT	0.8355	27
	DKVMN	0.8280	50
	DKT+	0.8274	62
	KQN	0.8230	16

On Junyi2015, All-as-Vision reaches its best validation AUC at epoch 6, earlier than simpleKT, stableKT, DKVMN, and KQN while also obtaining the strongest AUC in the table. On XES3G5M, the same pattern becomes clearer: All-as-Vision peaks at epoch 9, ahead of stableKT, simpleKT, DKT, DKVMN, and DKT+, again while remaining the strongest model by validation AUC. We do not claim that visual tracing always reaches the earliest checkpoint among all baselines, but the trend is consistent with the idea that a more structured representation reduces the burden placed on the tracing backbone during optimization.

6 Conclusion

We have presented *All-as-Vision*, a unified visual paradigm for KT that shifts the burden of generalization from sparse identity lookup to transferable exercise semantics. By coupling rendered exercise canvases with frozen visual embeddings and a lightweight gated temporal module, the framework remains competitive under standard benchmarking, becomes markedly stronger under strict unseen-question and unseen-concept evaluation, and supports zero-shot cross-dataset transfer through a representation interface not tied to dataset-local IDs. The broader implication is that semantic robustness in KT may depend less on endlessly enlarging sequence models than on giving them an input space whose local geometry already respects exercise meaning, notation, and visual structure. In this sense, All-as-Vision is not merely a modality substitution, but a redefinition of the exercise interface itself. Taken together, the results suggest that more generalizable KT may arise less from ever-heavier temporal architectures than from richer and more portable semantic representations from the outset.

References

[1] Sanjeev Arora, Hrishikesh Khandeparkar, Mikhail Khodak, Orestis Plevrakis, and Nikunj Saunshi. 2019. A Theoretical Analysis of Contrastive Unsupervised Representation Learning. arXiv:1902.09229 [cs.LG] https://arxiv.org/abs/1902.09229

[2] Youngduck Choi, Youngnam Lee, Junghyun Cho, Jineon Baek, Byungsoo Kim, Yeongmin Cha, Dongmin Shin, Chan Bae, and Jaewe Heo. 2020. Towards an Appropriate Query, Key, and Value Computation for Knowledge Tracing. In *Proceedings of the Seventh ACM Conference on Learning @ Scale* (Virtual Event, USA) *(L@S '20)*. Association for Computing Machinery, New York, NY, USA, 341–344. doi:10.1145/3386527.3405945

[3] Aritra Ghosh, Neil Heffernan, and Andrew S. Lan. 2020. Context-Aware Attentive Knowledge Tracing. arXiv:2007.12324 [cs.LG] https://arxiv.org/abs/2007.12324

[4] Xiaopeng Guo, Zhijie Huang, Jie Gao, Mingyu Shang, Maojing Shu, and Jun Sun. 2021. Enhancing Knowledge Tracing via Adversarial Training. arXiv:2108.04430 [cs.CY] https://arxiv.org/abs/2108.04430

[5] Jeff Z. HaoChen, Colin Wei, Adrien Gaidon, and Tengyu Ma. 2022. Provable Guarantees for Self-Supervised Deep Learning with Spectral Contrastive Loss. arXiv:2106.04156 [cs.LG] https://arxiv.org/abs/2106.04156

[6] Yoonjin Im, Eunseong Choi, Heejin Kook, and Jongwuk Lee. 2023. Forgetting-aware Linear Bias for Attentive Knowledge Tracing. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (CIKM '23)*. ACM, 3958–3962. doi:10.1145/3583780.3615191

[7] Junyi Academy Foundation. [n. d.]. Junyi Academy Online Learning Activity Dataset. Kaggle dataset. https://www.kaggle.com/datasets/junyiacademy/learning-activity-public-dataset-by-junyi-academy Accessed: 2026-03-28.

[8] Geewook Kim, Teakgyu Hong, Moonbin Yim, Jeongyeon Nam, Jinyoung Park, Jinyeong Yim, Wonseok Hwang, Sangdoo Yun, Dongyoon Han, and Seunghyun Park. 2022. OCR-free Document Understanding Transformer. arXiv:2111.15664 [cs.LG] https://arxiv.org/abs/2111.15664

[9] Chungpa Lee, Joonhwan Chang, and Jy yong Sohn. 2024. Analysis of Using Sigmoid Loss for Contrastive Learning. arXiv:2402.12613 [cs.LG] https://arxiv.org/abs/2402.12613

[10] Jinseok Lee and Dit-Yan Yeung. 2019. Knowledge Query Network for Knowledge Tracing: How Knowledge Interacts with Skills. In *Proceedings of the 9th International Conference on Learning Analytics & Knowledge (LAK19)*. ACM, 491–500. doi:10.1145/3303772.3303786

[11] Kenton Lee, Mandar Joshi, Iulia Turc, Hexiang Hu, Fangyu Liu, Julian Eisenschlos, Urvashi Khandelwal, Peter Shaw, Ming-Wei Chang, and Kristina Toutanova. 2023. Pix2Struct: Screenshot Parsing as Pretraining for Visual Language Understanding. arXiv:2210.03347 [cs.CL] https://arxiv.org/abs/2210.03347

[12] Meir Yossef Levi and Guy Gilboa. 2025. The Double-Ellipsoid Geometry of CLIP. arXiv:2411.14517 [cs.CV] https://arxiv.org/abs/2411.14517

[13] Xueyi Li, Youheng Bai, Teng Guo, Zitao Liu, Yaying Huang, Xiangyu Zhao, Feng Xia, Weiqi Luo, and Jian Weng. 2024. Enhancing Length Generalization for Attention Based Knowledge Tracing Models with Linear Biases. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI-24*, Kate Larson (Ed.). International Joint Conferences on Artificial Intelligence Organization, 5918–5926. doi:10.24963/ijcai.2024/654 Main Track.

[14] Qi Liu, Zhenya Huang, Yu Yin, Enhong Chen, Hui Xiong, Yu Su, and Guoping Hu. 2019. EKT: Exercise-aware Knowledge Tracing for Student Performance Prediction. arXiv:1906.05658 [cs.CY] https://arxiv.org/abs/1906.05658

[15] Yunfei Liu, Yang Yang, Xianyu Chen, Jian Shen, Haifeng Zhang, and Yong Yu. 2020. Improving Knowledge Tracing via Pre-training Question Embeddings. arXiv:2012.05031 [cs.IR] https://arxiv.org/abs/2012.05031

[16] Zitao Liu, Qiongqiong Liu, Jiahao Chen, Shuyan Huang, and Weiqi Luo. 2023. simpleKT: A Simple But Tough-to-Beat Baseline for Knowledge Tracing. arXiv:2302.06881 [cs.LG] https://arxiv.org/abs/2302.06881

[17] Zitao Liu, Qiongqiong Liu, Teng Guo, Jiahao Chen, Shuyan Huang, Xiangyu Zhao, Jiliang Tang, Weiqi Luo, and Jian Weng. 2023. XES3G5M: A Knowledge Tracing Benchmark Dataset with Auxiliary Information. In *Advances in Neural Information Processing Systems*, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (Eds.), Vol. 36. Curran Associates, Inc., 32958–32970. https://proceedings.neurips.cc/paper_files/paper/2023/file/67fc628f17c2ad53621fb961c6bafcaf-Paper-Datasets_and_Benchmarks.pdf

[18] Koki Nagatani, Qian Zhang, Masahiro Sato, Yan-Ying Chen, Francine Chen, and Tomoko Ohkuma. 2019. Augmenting Knowledge Tracing by Considering Forgetting Behavior. In *The World Wide Web Conference* (San Francisco, CA, USA) *(WWW '19)*. Association for Computing Machinery, New York, NY, USA, 3101–3107. doi:10.1145/3308558.3313565

[19] Shalini Pandey and George Karypis. 2019. A Self-Attentive model for Knowledge Tracing. arXiv:1907.06837 [cs.LG] https://arxiv.org/abs/1907.06837

[20] Chris Piech, Jonathan Spencer, Jonathan Huang, Surya Ganguli, Mehran Sahami, Leonidas J. Guibas, and Jascha Sohl-Dickstein. 2015. Deep Knowledge Tracing. *CoRR* abs/1506.05908 (2015). arXiv:1506.05908 http://arxiv.org/abs/1506.05908

[21] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning Transferable Visual Models From Natural Language Supervision. arXiv:2103.00020 [cs.CV] https://arxiv.org/abs/2103.00020

[22] Dongmin Shin, Yugeun Shim, Hangyeol Yu, Seewoo Lee, Byungsoo Kim, and Youngduck Choi. 2021. SAINT+: Integrating Temporal Features for EdNet Correctness Prediction. In *LAK21: 11th International Learning Analytics and Knowledge Conference (LAK21)*. ACM, 490–496. doi:10.1145/3448139.3448188

[23] Michael Tschannen, Alexey Gritsenko, Xiao Wang, Muhammad Ferjad Naeem, Ibrahim Alabdulmohsin, Nikhil Parthasarathy, Talfan Evans, Lucas Beyer, Ye Xia, Basil Mustafa, Olivier Hénaff, Jeremiah Harmsen, Andreas Steiner, and Xiaohua Zhai. 2025. SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features. arXiv:2502.14786 [cs.CV] https://arxiv.org/abs/2502.14786

[24] Tongzhou Wang and Phillip Isola. 2022. Understanding Contrastive Representation Learning through Alignment and Uniformity on the Hypersphere. arXiv:2005.10242 [cs.LG] https://arxiv.org/abs/2005.10242

[25] Wentao Wang, Huifang Ma, Yan Zhao, Fanyi Yang, and Liang Chang. 2022. SEEP: Semantic-enhanced question embeddings pre-training for improving knowledge tracing. *Information Sciences* 614 (2022), 153–169. doi:10.1016/j.ins.2022.10.015

[26] Zichao Wang, Angus Lamb, Evgeny Saveliev, Pashmina Cameron, Yordan Zaykov, José Miguel Hernández-Lobato, Richard E. Turner, Richard G. Baraniuk, Craig Barton, Simon Peyton Jones, Simon Woodhead, and Cheng Zhang. 2021. Instructions and Guide for Diagnostic Questions: The NeurIPS 2020 Education Challenge. arXiv:2007.12061 [cs.CY] https://arxiv.org/abs/2007.12061

[27] Haoran Wei, Yaofeng Sun, and Yukun Li. 2025. DeepSeek-OCR: Contexts Optical Compression. arXiv:2510.18234 [cs.CV] https://arxiv.org/abs/2510.18234

[28] Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. 2023. Sigmoid Loss for Language Image Pre-Training. arXiv:2303.15343 [cs.CV] https://arxiv.org/abs/2303.15343

[29] Dejiao Zhang, Feng Nan, Xiaokai Wei, Shangwen Li, Henghui Zhu, Kathleen McKeown, Ramesh Nallapati, Andrew Arnold, and Bing Xiang. 2021. Supporting Clustering with Contrastive Learning. arXiv:2103.12953 [cs.LG] https://arxiv.org/abs/2103.12953

[30] Jiani Zhang, Xingjian Shi, Irwin King, and Dit-Yan Yeung. 2017. Dynamic Key-Value Memory Networks for Knowledge Tracing. arXiv:1611.08108 [cs.AI] https://arxiv.org/abs/1611.08108